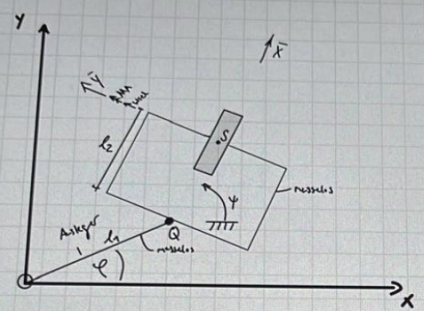
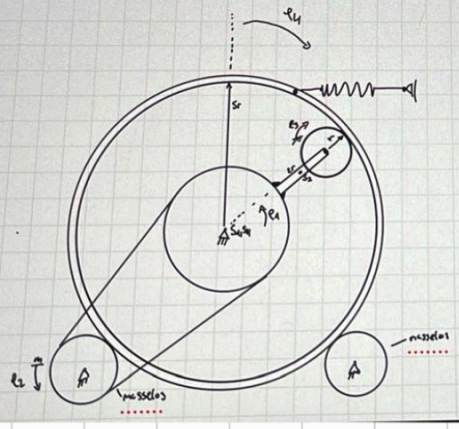
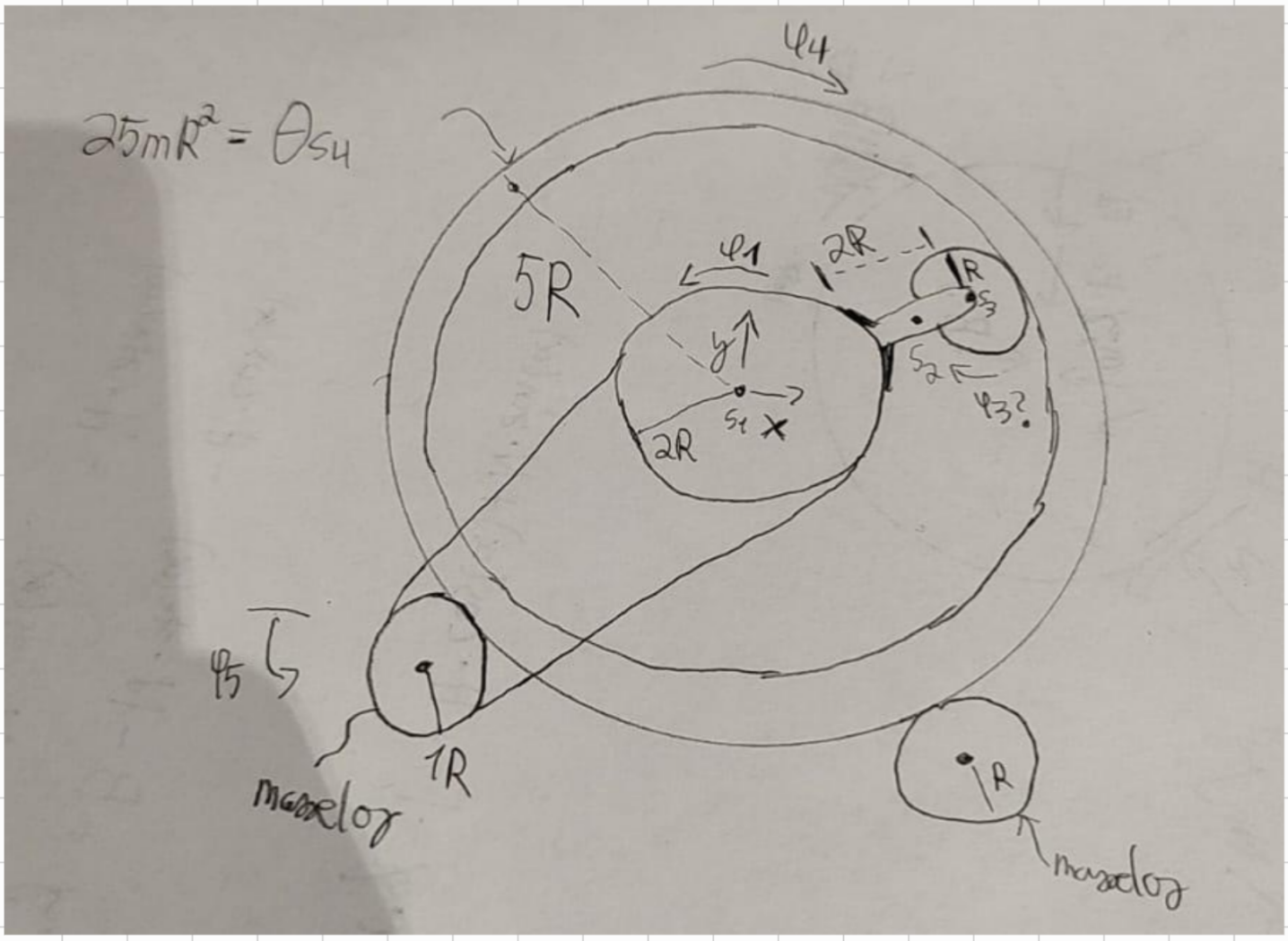
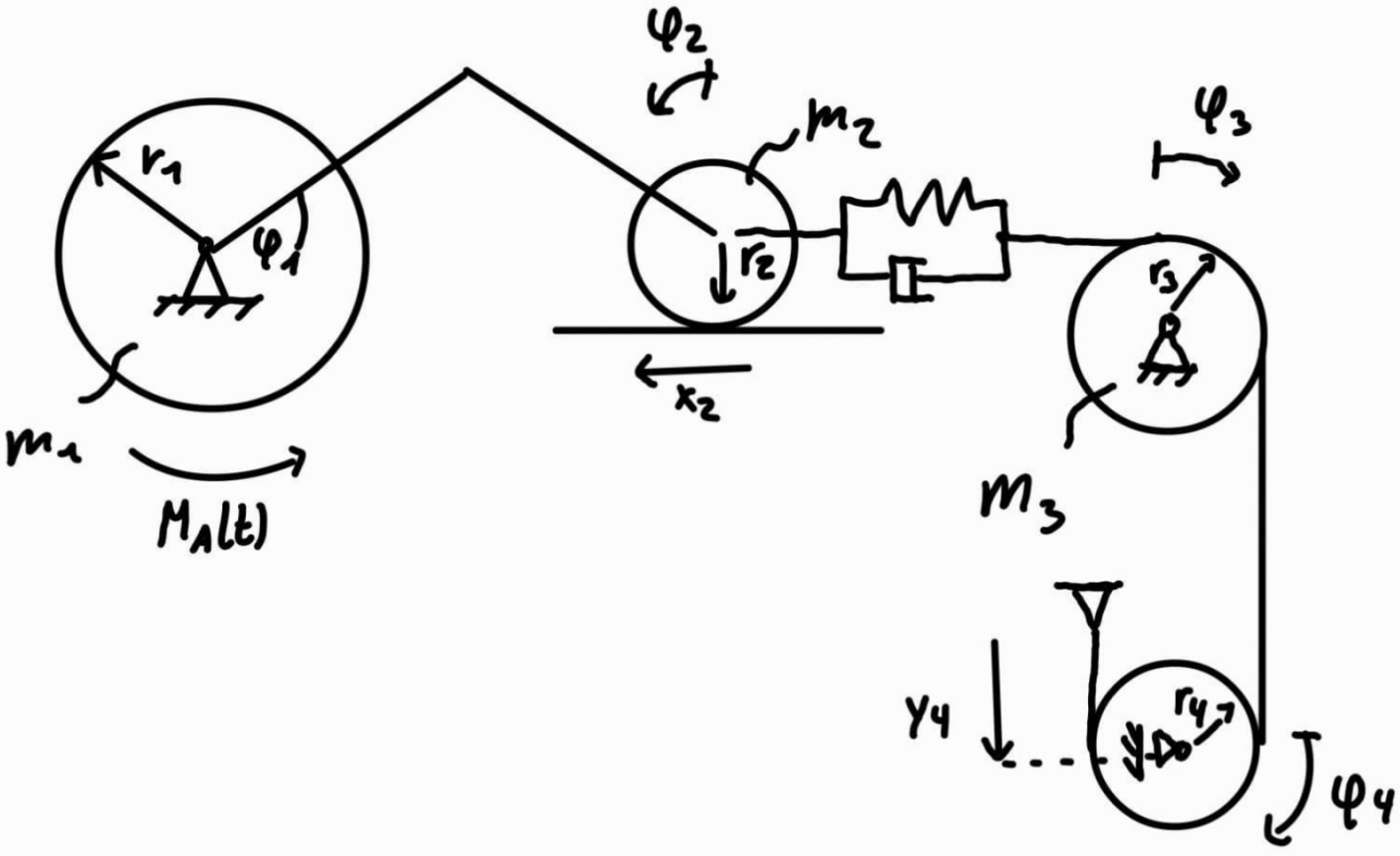
(a) Θ_Q (im $\vec{x},\vec{y},\vec{z}$ System) berechnen
(b) Θ_S (im $\vec{x},\vec{y},\vec{z}$ System) berechnen
(c) Schreibe freischneiden, Schwerpunktsatz und Drallsatz bzgl. S anwenden
(d) Bestimmen (ohne Schnitt und Auflager) freischneiden Kräfte in Q ermitteln





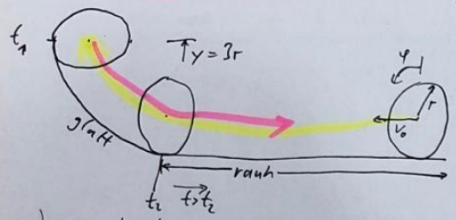



φ2
m2
φ3
r1
φ1
r2
r3
x2
m1
MA(t)
m3
y4
r4
φ4

TM3 WiSe 23/24 „die GOTTLOSE"

A1 rollt von rechts nach links los gleitet zyl Wand hoch bis t_1 ab t_2 beim Weg von rechts nach links wieder nach

KEINE Angabe zu φ_0
=> Muss aus v_0 & r hergeleitet werden für T_0



geg: $\mu_0 \overset{???}{=} \mu$

a) Wie hoch muss v_0 sein, damit die Walze in ihrer höchsten ~~t~~ Lage ohne „Geschwindigkeit" ankommt
UNKLAR: auch ohne Rotationsgeschw ???
lauft Aufgstellung klang es nach KOMPLETT ohne Geschw
(eig unlogisch weil Walze keinen Grund hat rotation aufzuhören)

b) (Annahme: bei t_1 keinerlei „Geschwindigkeit")
für $t \geq t_2$ Freikörperbild ohne Trägheiten aufstellen
und Schwerpunktsatz + Drallsatz → daraus durch Integration $\dot{\varphi}(t)$ und $\dot{x}(t)$ berechnen



$\circlearrowleft \quad \Theta\ddot{\varphi} = -rR = -r\mu N = -r\mu mg$

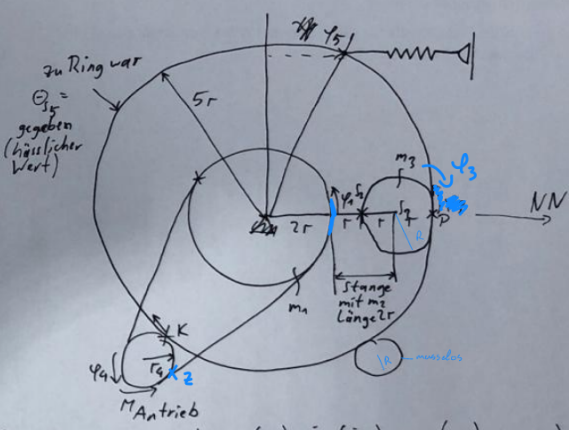
$\Theta = \frac{2}{5}mr^2 \quad \Rightarrow \ddot{\varphi} = \frac{1}{\frac{2}{5}mr^2}(-r\mu mg)$

c) d) e) Wann fängt Walze wieder an zu rollen?
Wie muss Radius gewählt ~~wird~~ werden damit Walze bei $x = ($ anfängt zu rollen?

A2 WiSe 23/24

Ausgangslage: $\varphi_1 = \varphi_3 = \varphi_4 = \varphi_5 = 0$
Die Feder ist so vorgespannt, dass sich das System entgegen der „Schwerkraft in positive φ_1 Richtung los bewegt"
„Feder entspannt bei: $\frac{\pi}{6} = \varphi_5$"





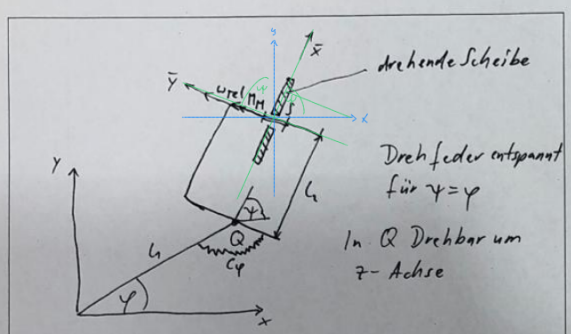
Berechnen: $\dot{\varphi}_4(\dot{\varphi}_1), \dot{\varphi}_3(\dot{\varphi}_1), \dot{r}_{S_2}(\dot{\varphi}_1), \varphi_5(\varphi_1), \dot{r}_{S_3}(\dot{\varphi}_1)$

$\dot{\varphi}_5 \cdot 5r = \dot{\varphi}_4 r_4 = 2r\dot{\varphi}_1$

Kin: $\dot{r}_P = \dot{\varphi}_5 \cdot 5r = \dot{r}_{P_{S_3}} + \dot{\varphi}_3 e_z \times \rho_{S_3 P}$

c) Aufstellen: $U(\varphi_1) \quad T(\dot{\varphi}_1)$
U_0 (nur Feder weil Rest im NN) $= \frac{1}{2}c\left(\sin\left(\varphi_5 - \frac{\pi}{6}\right)\right)^2$
$T_0 = 0$

A3 WiSe 23/24



a) a) Berechlennigung a_Q im $\bar{x}\bar{y}\bar{z}$-System
b) Berechlennigung a_S im $\bar{x}\bar{y}\bar{z}$-System
c) Freischnitt mit Schwerpunktsatz &



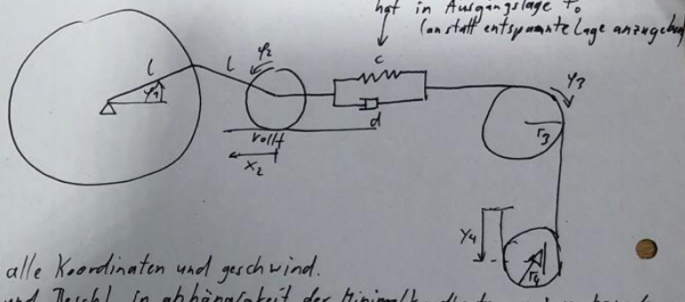
d) Freischnitt mit Schwerpunktsatz & Drallsatz um Lagerkräfte in Q zu berechnen

A4

Minimalkoord: y_4 & φ_1

keinerlei Θ gegeben
Alle Massebehaftet

hat im Ausgangs hat in Ausgangslage F_0 (anstatt entspannte Lage angezeigt)



a) alle Koordinaten und geschwind. und Beschl in abhängigkeit der Minimalkoordinaten y_4 & φ_1 bzw deren Ableitungen

b) Virtuelle Verrückungen δx_2

c) $U(y_1, y_4) \qquad T(\dot{\varphi}_1, \dot{y}_4)$

a) $T_1 \stackrel{!}{=} 0$

$T_0 = \frac{1}{2} m v_0^2 + \frac{1}{2} \Theta_1 \dot{\varphi}_0^2 = \frac{1}{2} m v_0^2 + \frac{1}{4} m r^2 \dot{\varphi}_0^2$ $\qquad \Theta_1 = \frac{1}{2} m r^2$

$U_1 = mgh = mg\,3r$

$U_0 \stackrel{!}{=} 0$

mit $\underline{v}_0 = \underline{v}_m + \begin{pmatrix} 0 \\ 0 \\ \dot{\varphi}_0 \end{pmatrix} \times \begin{pmatrix} 0 \\ r \\ 0 \end{pmatrix} = -r\dot{\varphi}_0 \quad \Rightarrow \quad \dot{\varphi}_0 = -\frac{v_0}{r}$

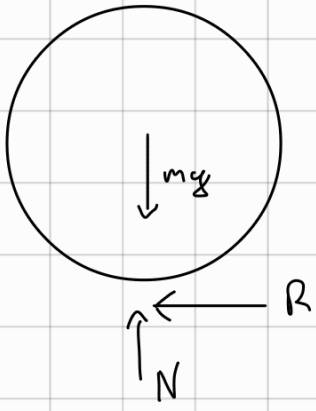
$T_0 = U_1$

$\frac{1}{2} m v_0^2 + \frac{1}{4} m r^2 \dot{\varphi}_0^2 = mg(3r - r)$

$\frac{1}{2} m v_0^2 + \frac{1}{4} m r^2 \frac{v_0^2}{r^2} = mg(3r - r) \quad \Rightarrow \quad \frac{3}{4} m v_0^2 = mg\,2r$

$\Rightarrow v_0 = \sqrt{\frac{8}{3} gr}$

b)



$m\ddot{x} = -R = -\mu mg$

$m\ddot{y} = N - mg \stackrel{!}{=} 0 \quad \Rightarrow N = mg$

$\Theta_{S_1} \ddot{\varphi} = R \cdot r = \mu mgr = \frac{1}{2} m r^2 \ddot{\varphi}$

mit $R = \mu N = \mu mg$

$\dot{\varphi}(t)$: $\quad \frac{1}{2} m r^2 \ddot{\varphi} = \mu mgr \quad \Rightarrow \quad \ddot{\varphi} = \frac{2\mu mgr}{mr^2} = \frac{2\mu g}{r}$

$\frac{d\dot{\varphi}}{dt} = \frac{2\mu g}{r}$

$\int 1 \, d\dot{\varphi} = \int \frac{2\mu g}{r} \, dt$

$\dot{\varphi} + C_1 = \frac{2\mu g}{r} t + C_2$

$\Rightarrow \quad C_2 - C_1 = C \qquad \dot{\varphi}(t) = \frac{2\mu g}{r} t + C$

$\dot{\varphi}(t_1) \overset{!}{=} 0 \quad \Rightarrow \quad C = -\frac{2\mu g}{r} t_1$

$\Rightarrow \dot{\varphi}(t) = \frac{2\mu g}{r}(t - t_1)$

$\dot{x}(t)$:

$\ddot{x} = -\mu g$

$\frac{d\dot{x}}{dt} = -\mu g$

$\int 1 \, d\dot{x} = \int -\mu g \, dt \qquad C_4 - C_3 = C$

$\dot{x}(t) = -\mu g t + C$

$\dot{x}(t_1) \overset{!}{=} 0 \quad \Rightarrow \quad C = \mu g t_1$

$\Rightarrow \quad \dot{x}(t) = \mu g (t_1 - t)$

c)

$t \overset{!}{=} t_2 \qquad 2$

d)

$x = \int \mu g t_1 - \mu g t \, dt$

A2

a)

$$\underline{v}_z = \underline{v}_{s4}^{\,0} + \begin{pmatrix}0\\0\\\dot\varphi_4\end{pmatrix} \times \begin{pmatrix}r\\0\\0\end{pmatrix} = r\dot\varphi_4$$

$$\underline{v}_z = \underline{v}_{s1}^{\,0} + \begin{pmatrix}0\\0\\\dot\varphi_1\end{pmatrix} \times \begin{pmatrix}2r\\0\\0\end{pmatrix} = 2r\dot\varphi_1$$

$$\underline{v}_z \doteq \underline{v}_z \qquad r\dot\varphi_4 = 2r\dot\varphi_1 \qquad \Rightarrow \qquad \boxed{\dot\varphi_4 = 2\dot\varphi_1}$$

$$\underline{v}_{s3} = \underline{v}_{s1}^{\,0} + \begin{pmatrix}0\\0\\\dot\varphi_1\end{pmatrix} \times \begin{pmatrix}4r\\0\\0\end{pmatrix} = \begin{pmatrix}0\\\dot\varphi_1 4r\\0\end{pmatrix} \qquad \boxed{\underline{v}_{s3} = +4r\dot\varphi_1}$$

$$\underline{v}_{s2} = \underline{v}_{s1}^{\,0} + \begin{pmatrix}0\\0\\\dot\varphi_1\end{pmatrix} \times \begin{pmatrix}3r\\0\\0\end{pmatrix} = \begin{pmatrix}0\\3r\dot\varphi_1\\0\end{pmatrix} \qquad \boxed{\underline{v}_{s_2} = 3r\dot\varphi_1}$$

$$\underline{v}_k = \underline{v}_{s4}^{\,0} + \begin{pmatrix}0\\0\\\dot\varphi_4\end{pmatrix} \times \begin{pmatrix}r\\0\\0\end{pmatrix} = \begin{pmatrix}0\\r\dot\varphi_4\\0\end{pmatrix}$$

$$-\underline{v}_p = \underline{v}_{s3} + \begin{pmatrix}0\\0\\-\dot\varphi_3\end{pmatrix} \times \begin{pmatrix}r\\0\\0\end{pmatrix} = \begin{pmatrix}0\\4r\dot\varphi_1\\0\end{pmatrix} + \begin{pmatrix}0\\-r\dot\varphi_3\\0\end{pmatrix}$$

$$\underline{v}_k \doteq -\underline{v}_p \qquad r\dot\varphi_4 = -\left(4r\dot\varphi_1 - r\dot\varphi_3\right)$$

$$2r\dot\varphi_1 + 4r\dot\varphi_1 = r\dot\varphi_3$$

$$\Rightarrow \quad \dot\varphi_3 = 6\dot\varphi_1$$

$$\underline{v}_k = \underline{v}_{s1}^{\,0} + \begin{pmatrix}0\\0\\-\dot\varphi_5\end{pmatrix} \times \begin{pmatrix}-5r\\0\\0\end{pmatrix} = \begin{pmatrix}0\\5r\dot\varphi_5\\0\end{pmatrix}$$

$$\underline{v}_k \doteq \underline{v}_k$$

$$r\dot\varphi_4 = 5r\dot\varphi_5 \qquad \Rightarrow \dot\varphi_5 = \frac{2}{5}\dot\varphi_1$$

$$\varphi_5 = \frac{2}{5}\varphi_1$$

c)

$$T = \frac{1}{2}\Theta_{s_1}\dot{\varphi}_1^2 + \frac{1}{2}m_2\dot{r}_{s_2}^2 + \frac{1}{2}\Theta_{s_2}\dot{\varphi}_1^2 + \frac{1}{2}m_3\dot{r}_{s_3}^2 + \frac{1}{2}\Theta_{s_3}\dot{\varphi}_3^2 + \frac{1}{2}\Theta_{s_5}\dot{\varphi}_5^2$$

$$\Theta_{s_1} = \frac{1}{2}m(2r)^2 = 2m_1r^2$$

$$\Theta_{s_2} = \frac{1}{12}m_2(2r)^2 = \frac{1}{3}m_2r^2$$

$$\Theta_{s_3} = \frac{1}{2}m_3r^2 = \frac{1}{2}m_3r^2$$

$$\Theta_{s_5} = 25m_5r^2$$

$$\Rightarrow T = m_1r^2\dot{\varphi}_1^2 + \frac{9}{2}m_2r^2\dot{\varphi}_1^2 + \frac{1}{6}m_2r^2\dot{\varphi}_1^2 + 8m_3r^2\dot{\varphi}_1^2 + 9m_3r^2\dot{\varphi}_1^2 + 2m_5r^2\dot{\varphi}_1^2$$

$$U = m_2g\cdot\sin(\varphi_1)\cdot 3r + m_3g\sin(\varphi_1)\cdot 4r + \frac{1}{2}c\left[5r\sin(\varphi_5) - 5r\sin\left(\frac{\pi}{6}\right)\right]^2$$

$$T_0 = 0 \qquad U_0 = \frac{1}{2}c\,25r^2\sin^2\left(\frac{\pi}{6}\right)$$

$$25r^2\sin^2(\varphi_5) + 25r^2\sin^2\left(\frac{\pi}{6}\right)$$

$$-50r^2\sin(\varphi_5)\sin\left(\frac{\pi}{6}\right)$$

d)

$$T_1 + U_1 = T_0 + U_0$$

$$\dot{\varphi}_1^2\left[m_1r^2 + \frac{15}{3}m_2r^2 + 17m_3r^2 + 2m_5r^2\right] = \frac{25}{2}cr^2\sin^2\left(\frac{\pi}{6}\right) - m_2g\sin(\varphi_1)3r - m_3g\sin(\varphi_1)4r - \frac{1}{2}c\left[5r\sin\left(\frac{2}{5}\varphi_1\right) - 5r\sin\left(\frac{\pi}{6}\right)\right]^2$$

$$\dot{\varphi}_1 = \frac{\frac{25}{2}cr^2\sin^2\left(\frac{\pi}{6}\right) - m_2g\sin(\varphi_1)3r - m_3g\sin(\varphi_1)4r - \frac{1}{2}c\left[5r\sin\left(\frac{2}{5}\varphi_1\right) - 5r\sin\left(\frac{\pi}{6}\right)\right]^2}{\left[m_1r^2 + \frac{15}{3}m_2r^2 + 17m_3r^2 + 2m_5r^2\right]}$$

A3

a)

x, y, z

$$\underline{r}_Q = \begin{pmatrix} l_1 \cos(\varphi) \\ l_1 \sin(\varphi) \\ 0 \end{pmatrix}$$

Transformation: $\underline{e}_{\bar{x}} = \cos(\psi)\,\underline{e}_x + \sin(\psi)\,\underline{e}_y$

$\underline{e}_{\bar{y}} = -\sin(\psi)\,\underline{e}_x + \cos(\psi)\,\underline{e}_y$

$$\dot{\underline{r}}_Q = \begin{pmatrix} -l_1 \dot{\varphi} \sin(\varphi) \\ l_1 \dot{\varphi} \cos(\varphi) \\ 0 \end{pmatrix}$$

$$\ddot{\underline{r}}_Q = \begin{pmatrix} -l_1 \ddot{\varphi} \sin(\varphi) - l_1 \dot{\varphi}^2 \cos(\varphi) \\ l_1 \ddot{\varphi} \cos(\varphi) - l_1 \dot{\varphi}^2 \sin(\varphi) \\ 0 \end{pmatrix}$$

$\bar{x}, \bar{y}, \bar{z}$

$$\ddot{\underline{r}}_Q = \begin{pmatrix} \left[-l_1 \ddot{\varphi} \sin(\varphi) - l_1 \dot{\varphi}^2 \cos(\varphi)\right] \cos(\psi) + \left[l_1 \ddot{\varphi} \cos(\varphi) - l_1 \dot{\varphi}^2 \sin(\varphi)\right] \sin(\psi) \\ -\left[-l_1 \ddot{\varphi} \sin(\varphi) - l_1 \dot{\varphi}^2 \cos(\varphi)\right] \sin(\psi) + \left[l_1 \ddot{\varphi} \cos(\varphi) - l_1 \dot{\varphi}^2 \sin(\varphi)\right] \cos(\psi) \\ 0 \end{pmatrix}$$

b)

x, y, z

$$\underline{r}_S = \begin{pmatrix} l_1 \cos(\varphi) + l_2 \cos(\psi) \\ l_1 \sin(\varphi) + l_2 \sin(\varphi) \\ 0 \end{pmatrix}$$

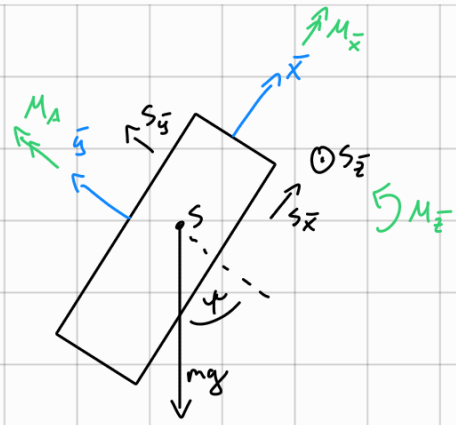
$$\dot{\underline{r}}_S = \begin{pmatrix} -l_1 \dot{\varphi} \sin(\varphi) - l_2 \dot{\psi} \sin(\psi) \\ l_1 \dot{\varphi} \cos(\varphi) + l_2 \dot{\psi} \cos(\psi) \\ 0 \end{pmatrix}$$

$$\ddot{\underline{r}}_S = \begin{pmatrix} -l_1 \ddot{\varphi} \sin(\varphi) - l_1 \dot{\varphi}^2 \cos(\varphi) - l_2 \ddot{\psi} \sin(\psi) - l_2 \dot{\psi}^2 \cos(\psi) \\ l_1 \ddot{\varphi} \cos(\varphi) - l_1 \dot{\varphi}^2 \sin(\varphi) + l_2 \ddot{\psi} \cos(\psi) - l_2 \dot{\psi}^2 \sin(\psi) \\ 0 \end{pmatrix}$$

in $\bar{x}, \bar{y}, \bar{z}$

$$\ddot{\underline{r}}_S = \begin{pmatrix} \left[-l_1 \ddot{\varphi} \sin(\varphi) - l_1 \dot{\varphi}^2 \cos(\varphi) - l_2 \ddot{\psi} \sin(\psi) - l_2 \dot{\psi}^2 \cos(\psi)\right] \cos(\psi) + \left[l_1 \ddot{\varphi} \cos(\varphi) - l_1 \dot{\varphi}^2 \sin(\varphi) + l_2 \ddot{\psi} \cos(\psi) - l_2 \dot{\psi}^2 \sin(\psi)\right] \sin(\psi) \\ -\left[-l_1 \ddot{\varphi} \sin(\varphi) - l_1 \dot{\varphi}^2 \cos(\varphi) - l_2 \ddot{\psi} \sin(\psi) - l_2 \dot{\psi}^2 \cos(\psi)\right] \sin(\psi) + \left[l_1 \ddot{\varphi} \cos(\varphi) - l_1 \dot{\varphi}^2 \sin(\varphi) + l_2 \ddot{\psi} \cos(\psi) - l_2 \dot{\psi}^2 \sin(\psi)\right] \cos(\psi) \\ 0 \end{pmatrix}$$

c)



$$m\ddot{x}_S = S_{\bar{x}} - mg\sin(\psi)$$
$$m\ddot{y}_S = S_{\bar{y}} - mg\cos(\psi)$$
$$m\ddot{z}_S = S_{\bar{z}} = 0$$

Annahme Θ_{S_1}, $\Theta_{S_2} = \Theta_{S_3}$

$$\Theta_{S_1}\dot{\omega}_{\bar{x}} - \left[\Theta_{S_2}\omega_{\bar{y}}\Omega_{\bar{z}} - \Theta_{S_3}\omega_{\bar{z}}\Omega_{\bar{y}}\right] = M_{S\bar{x}}$$

$$\Theta_{S_2}\dot{\omega}_{\bar{y}} - \left[\Theta_{S_3}\omega_{\bar{z}}\Omega_{\bar{x}} - \Theta_{S_1}\omega_{\bar{x}}\Omega_{\bar{z}}\right] = M_{S\bar{y}} = M_A$$

$$\Theta_{S_3}\dot{\omega}_{\bar{z}} - \left[\Theta_{S_1}\omega_{\bar{x}}\Omega_{\bar{y}} - \Theta_{S_2}\omega_{\bar{y}}\Omega_{\bar{x}}\right] = M_{S\bar{z}}$$

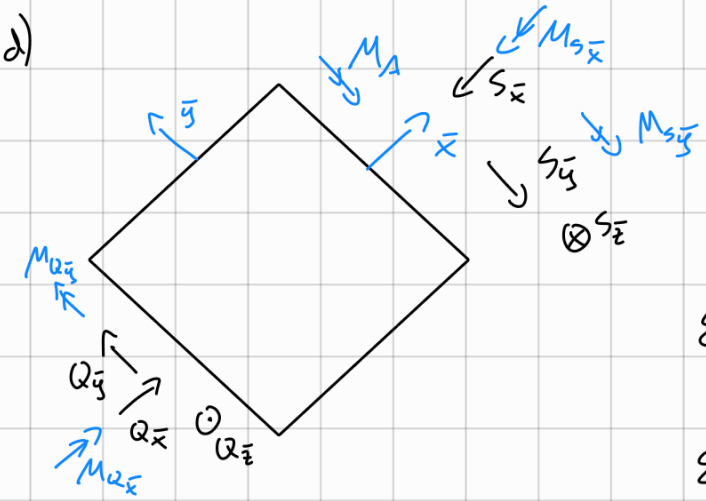
$$\underline{\omega} = \begin{pmatrix} 0 \\ \omega_{sel} \\ \dot{\psi} \end{pmatrix} \qquad \underline{\dot{\omega}} = \begin{pmatrix} 0 \\ \dot{\omega}_{sel} \\ \ddot{\psi} \end{pmatrix} \qquad \underline{\Omega} = \begin{pmatrix} 0 \\ 0 \\ \dot{\psi} \end{pmatrix}$$

$$-\Theta_{S_2}\omega_{\bar{y}}\Omega_{\bar{z}} = M_{S\bar{x}} \quad \Rightarrow \quad -\Theta_{S_2}\omega_{sel}\dot{\psi} = M_{S\bar{x}}$$

$$\Theta_{S_2}\dot{\omega}_{sel} = M_{S\bar{y}} = M_A \qquad \Theta_{S_2}\dot{\omega}_{sel} = M_A$$

$$\Theta_{S_3}\dot{\omega}_{\bar{z}} = M_{S\bar{z}} \qquad \Theta_{S_3}\ddot{\psi} = M_{S\bar{z}}$$

d)



$$\Sigma F_{\bar{x}} \stackrel{!}{=} 0: \quad Q_{\bar{x}} - S_{\bar{x}} = 0$$

$$\Sigma F_{\bar{y}} \stackrel{!}{=} 0: \quad Q_{\bar{y}} - S_{\bar{y}} = 0$$

$$\Sigma F_{\bar{z}} \stackrel{!}{=} 0: \quad Q_{\bar{z}} - S_{\bar{z}} = 0$$

$$\Sigma M_Q^{\bar{x}} \stackrel{!}{=} 0: \quad M_{Q\bar{x}} - M_{S\bar{x}} = 0$$

$$\Sigma M_Q^{\bar{y}} \stackrel{!}{=} 0: \quad M_{Q\bar{y}} - M_A + S_{\bar{z}} \cdot l_2 = 0$$

$$Q_{\bar{x}} = S_{\bar{x}} = m\ddot{x}_S + mg\sin(\varphi)$$

$$Q_{\bar{y}} = S_{\bar{y}} = m\ddot{y}_S + mg\cos(\varphi)$$

$$Q_{\bar{z}} = S_{\bar{z}} = 0$$

$$M_{Q\bar{x}} = M_{S\bar{x}} = -\Theta_{S_2}\omega_S \dot{\varphi}$$

$$M_{Q\bar{y}} = \Theta_{S_2}\dot{\omega}_S$$

(4)

$$\underline{r}_{S_1} = \begin{pmatrix} 2l \\ 0 \\ 0 \end{pmatrix} \qquad \underline{\dot{r}}_{S_1} = 0 \qquad \underline{\ddot{r}}_{S_1} = 0$$

$$\underline{r}_{S_2} = \begin{pmatrix} x_2 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2l - 2l\cos(\varphi_1) \\ 0 \\ 0 \end{pmatrix} \quad \underline{\dot{r}}_{S_2} = \begin{pmatrix} 2l\dot{\varphi}_1\sin(\varphi_1) \\ 0 \\ 0 \end{pmatrix} \quad \underline{\ddot{r}}_{S_2} = \begin{pmatrix} 2l\ddot{\varphi}_1\sin(\varphi_1) + 2l\dot{\varphi}_1^2\cos(\varphi_1) \\ 0 \\ 0 \end{pmatrix}$$

$$\underline{\dot{r}}_{S_4} = \begin{pmatrix} 0 \\ \dot{y}_4 \\ 0 \end{pmatrix} \qquad \underline{r}_{S_4} = \begin{pmatrix} 0 \\ y_4 \\ 0 \end{pmatrix} \qquad \underline{\ddot{r}}_{S_4} = \begin{pmatrix} 0 \\ \ddot{y}_4 \\ 0 \end{pmatrix}$$

b)

$$\delta r_{S_4} = \delta y_4 \qquad \delta r_{S_2} = 2l\sin(\varphi_1)\,\delta\varphi_1$$

c)

$$T = T_{S_1} + T_{S_2} + T_{S_3} + T_{S_4}$$

$$T_{S_1} = \frac{1}{2}\Theta_{S_1}$$